Chapter 12

Solids and modern materials

1. A solid has a very high melting point, great hardness, and poor electrical conduction. This is a(n) ____D____ solid.

A) ionic
B) molecular
C) metallic
D) covalent network
E) metallic and covalent network

2. Crystalline solids differ from amorphous solids in that crystalline solids have ____B____.

A) appreciable intermolecular attractive forces
B) a long-range repeating pattern of atoms, molecules, or ions
C) atoms, molecules, or ions that are close together
D) much larger atoms, molecules, or ions
E) no orderly structure

3. Which of the following cannot form a solid with a lattice based on the sodium chloride structure? ____DE____

A) NaBr
B) LiF
C) RbI
D) CuO
E) $CuCl_2$

Structure and physical properties [edit]

Copper(II) oxide belongs to the monoclinic crystal system. The copper atom is coordinated by 4 oxygen atoms in an approximately square planar configuration.[1]

4. Consider the following statements about crystalline solids, which of the statements is false? ____C____

A) Molecules or atoms in molecular solids are held together via intermolecular forces.
B) Metallic solids have atoms in the points of the crystal lattice.
C) Ionic solids have formula units in the point of the crystal lattice.
D) Atoms in covalent-network solids are connected via a network of covalent bonds.

5. CsCl crystallizes in a unit cell that contains the Cs^+ ion at the center of a cube that has a Cl^- at each corner. Each unit cell contains _________ Cs^+ ion(s) and ____C____ Cl^- ion(s), respectively.

A) 1 and 8
B) 2 and 1
C) 1 and 1
D) 2 and 2
E) 2 and 4

6. An alloy is a ____C____.

A) heterogeneous mixture of two metals.
B) pure metal.
C) metallic material that is composed of two or more elements.
D) nonmetal with some properties of a metal.
E) mineral containing two or more metals.

7. For a substitutional alloy to form, the two metals combined must have similar___D___.

A) ionization potential and electron affinity.
B) number of valance electrons and electronegativity.
C) reduction potential and size.
D) atomic radii and chemical bonding properties.
E) band gap and reactivity.

normally occupied by a solvent atom. Two metallic components have similar atomic radii and chemical-bonding characteristics

8. What is the typical effect of the addition of an interstitial element on the properties of a metal? __B____

A) increase in malleability and corrosion resistance
B) increase in hardness and strength, decrease in ductility
C) decrease in melting point and increase in ductility
D) decrease in conductivity and increase in brittleness
E) increased surface luster

9. Which will have a larger band gap: GaN, GaP, GaAs, or GaSb?

A
____

A) GaN

B) GaP

C) GaAs

D) GaSb

E) All choices have identical band gaps.

transition from pure covalent bonding in elemental semiconductors to polar covalent bonding in compound semiconductors. As the difference in electronegativity of the elements increases, the bonding becomes more polar and the band gap increases.

10. The _A____ lattice is one of the seven primitive three-dimensional lattices in which the relationship between the lattice vectors a, b, and c can be written as: $a \neq b \neq c$

A) monoclinic
B) hexagonal
C) rhombohedral
D) tetragonal
E) cubic

11. Gallium crystallizes in a primitive cubic unit cell. The length of the unit cell edge is 3.70 Å. The radius of a Ga atom is ____C____ Å.

A) 7.40
B) 3.70
C) 1.85
D) 0.930
E) Insufficient data is given.